SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – COMPARITIVE ANALYSIS

Course Code:	ITA0302	**Course Title:**	Mobile computing
CO Assessed:	CO4	**Assessment:**	AT1 - COMPARITIVE ANALYSIS
Weightage:	30 % of CIA	**Total Marks:**	100
CO4 Statement:	*Identify the components and structure of mobile application for Android and Windows OS-based devices.*		
Student Name:	DINDU PUJITHA	**Reg. No.:**	192373037
Section / Batch:	Slot D	**Date:**	25/08/2026

Code of Conduct

I DINDU PUJITHA(192373037) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: D PUJITHA

Instructions

- Duration: 60 minutes
- Number of questions : 01

Annexure A - QUESTIONS

1. **A university plans to provide smartphones to students and wants a platform that supports open-source development, extensive customization, and installation of applications from multiple sources. Compare Android and iOS architectures and development models. Determine which OS better satisfies these requirements and explain why.**

Android Architecture and Development Model:

Android is based on the Linux kernel and uses an open-source platform called the Android Open Source Project (AOSP). Its architecture consists of the Linux kernel, hardware abstraction layer, native libraries, Android Runtime (ART), application framework, and applications.

Android provides developers with a high level of flexibility. Developers can customize the operating system, modify system components, and create applications using tools such as Android Studio, Java, and Kotlin. Android also allows applications to be installed from the Google Play Store as well as third-party sources, commonly known as sideloading.

iOS Architecture and Development Model:

iOS is developed and controlled by Apple. It uses a layered architecture based on technologies such as the XNU kernel, system frameworks, and Apple's application environment. Unlike Android, iOS is closed-source, so developers cannot freely modify the operating system.

iOS application development mainly uses Xcode, Swift, and Objective-C. Applications are primarily distributed through the Apple App Store, and Apple applies strict security and approval policies. Installation from sources outside Apple's normal distribution mechanisms is highly restricted.

Comparison:

Feature	Android	Ios
Source model	Open-source core (AOSP)	Closed-source
Customization	High	Limited
Application development	Flexible	Controlled by Apple
Multiple app sources	Supported	Highly restricted
Sideloading	Supported	Generally restricted
Hardware/software flexibility	High	Limited to Apple devices
Developer control	High	More restricted

Suitable Operating System:

Android is the better choice for the university. Its open-source foundation supports development and customization, while its flexible architecture allows universities and developers to modify software according to their requirements. Android also supports application installation from multiple sources, which is important if the university wants to distribute its own educational or research applications outside the standard app store.

Conclusion:

Therefore, Android better satisfies the university's requirements because it provides an open-source development environment, extensive customization options, and greater freedom to install applications from different sources. iOS provides strong security and a controlled ecosystem, but its closed architecture and strict application distribution policies make it less suitable for these specific requirements.